

CHAPTER 3

Force, Mass & Motion

Why does a soccer ball roll? Why do you fly forward when the bus stops fast? Let's find out!



Easy Read Edition

WHAT WE WILL LEARN

- ★ What is motion?
- ★ Speed and direction
- ★ Speeding up and slowing down
- ★ What are forces?
- ★ Newton's First Law
- ★ Newton's Second Law
- ★ Newton's Third Law

3.1 What Is Motion?

Motion means something is moving. It changes its place over time.

But "moving" is tricky. Right now you might feel like you are sitting still. But:

- Earth is spinning at 1,000 miles per hour!
- Earth is flying around the Sun at 67,000 mph!
- You are zooming through space without knowing it!

So are you moving or not? **It depends on what you compare yourself to.**

That thing you compare to is called a **reference point**. If you compare yourself to your desk, you're not moving. If you compare to the Sun, you're flying!

3.2 Speed

Speed tells you how fast something moves. Here's the simple formula:

$$\text{Speed} = \text{Distance} \div \text{Time}$$

If you walk 100 meters in 20 seconds, your speed is 5 meters per second.



Walking
~3 mph



Biking
~15 mph



Car
~60 mph

Velocity = Speed + Direction

Velocity is speed PLUS direction.

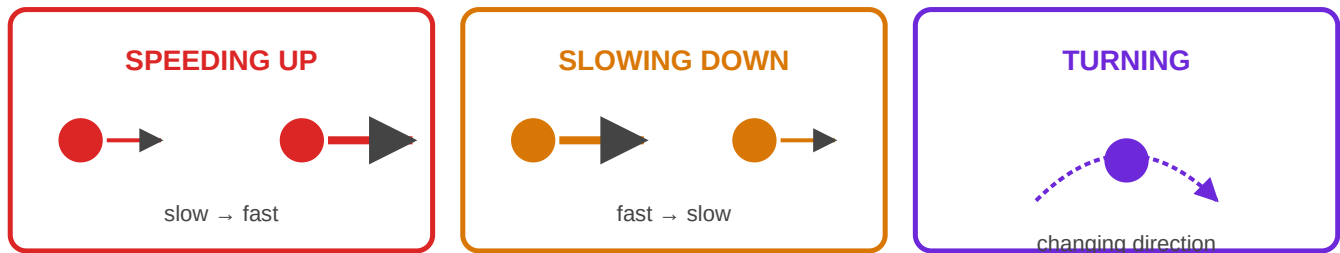
- "60 mph" is speed.
- "60 mph going north" is velocity.

3.3 Acceleration

Acceleration means velocity is changing. In real life, that means:

- **Speeding up** = acceleration
- **Slowing down** = acceleration (also called deceleration)
- **Changing direction** = also acceleration!

Three Kinds of Acceleration



All three of these count as acceleration!

3.4 Forces

A **force** is a push or a pull.

Forces can:

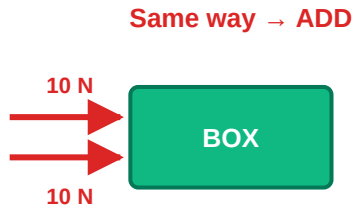
- Start something moving
- Stop something moving
- Speed it up or slow it down
- Change its direction

Forces are measured in **newtons (N)**. They are named for Isaac Newton — the scientist who figured this all out 300 years ago.

Net Force

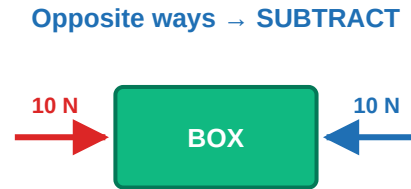
Usually more than one force pushes on something at the same time. The **net force** is the TOTAL force after you add them up.

Adding Up Forces



Net force = 20 N →

Box moves!



Net force = 0 N

Box doesn't move

Forces in the same direction ADD. Opposite forces SUBTRACT.

Balanced Forces

Forces cancel out. Net force = 0. The object does not change motion. It stays at rest, or keeps moving the same.

Unbalanced Forces

Forces don't cancel. Net force is NOT zero. The object's motion changes!

3.5 Newton's First Law

1 Law of Inertia

An object at rest stays at rest. An object in motion stays in motion. Unless a force changes it.

In other words: things keep doing what they're doing. They don't want to change.

Inertia is the word for this. It means resisting change in motion.

The more mass something has, the more inertia it has.

★ Inertia in Real Life

- School bus brakes hard → YOU fly forward (you wanted to keep moving!)
- Pulling a tablecloth fast → dishes stay (they don't want to move)
- Phone in your bag flies out → it wanted to keep going
- That's why we wear seat belts!

3.6 Newton's Second Law

2 $F = m \times a$

Force equals mass times acceleration.

This law tells you TWO things:

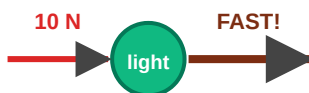
1. More force = more acceleration

Push something harder, and it speeds up faster.

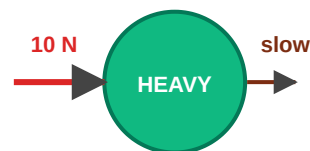
2. More mass = less acceleration

Bigger objects are harder to speed up. To move a bigger object, you need more force.

Same Push, Different Mass



light box → speeds up a lot



heavy box → barely moves

Same push but different mass = different acceleration.

That's why it's easy to push a shopping cart. But really hard to push a car!

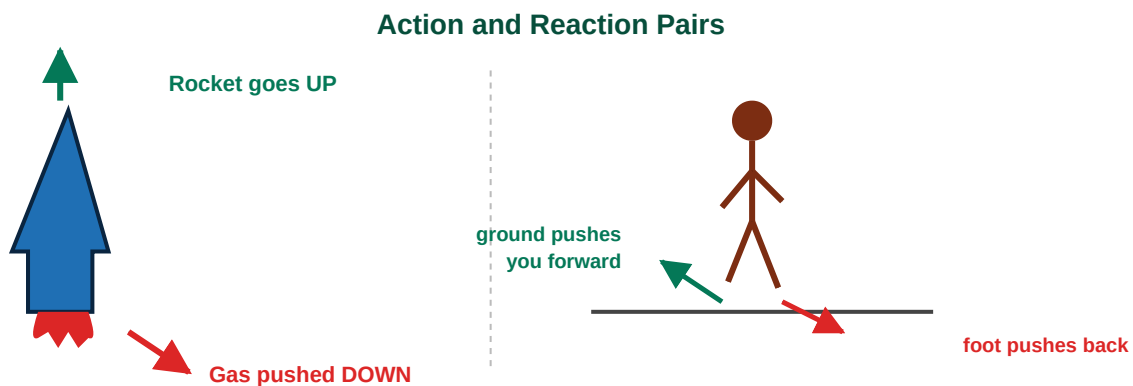
3.7 Newton's Third Law

3 Action and Reaction

For every action, there is an equal and opposite reaction.

Forces always come in **pairs**.

If you push something, it pushes you back the same amount.



Action	Reaction
You push the ground back when walking	Ground pushes you forward
Rocket pushes gas down	Gas pushes rocket up
Swimmer pushes water back	Water pushes swimmer forward
Bat hits ball	Ball pushes back on bat



Watch Out!

You might wonder: if forces come in pairs, why does anything move?

Because the two forces act on **different things**! When you push a wall, your force is on the wall. The wall's force is on you. They don't cancel each other.



Check Yourself

1. If you push a wall with 30 N, how much force does the wall push you with?
2. Why is it easier to push a tennis ball than a bowling ball with the same effort?
3. Why do you keep moving forward when the bus stops fast?

★ CHAPTER 3 — BIG IDEAS

- **Motion** is a change in place over time.
- **Speed** = distance ÷ time. **Velocity** = speed with a direction.
- **Acceleration** = any change in velocity (faster, slower, OR turning).
- A **force** is a push or pull. Measured in **newtons**.
- **Balanced** forces don't change motion. **Unbalanced** forces do.
- **Newton's 1st Law**: Things keep doing what they're doing (inertia).
- **Newton's 2nd Law**: $F = m \times a$. More mass = need more force.
- **Newton's 3rd Law**: Every action has an equal and opposite reaction.